Manuel Bonilla López, PhD

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LinkedIn Profile | Google Scholar | APS News Feature | Nature Nanotechnology Editorial

PROFESSIONAL SUMMARY

Lead-level manufacturing, process and systems engineer with a Ph.D. in Physics and 16+ years of experience across advanced manufacturing, semiconductors, microelectronics, optics, thin films, materials science and data analytics. Combines deep hands-on technical credibility with enterprise-scale systems leadership.

Recognized for solving high-priority operational problems across departments by recovering stalled implementations, stabilizing manufacturing processes, building SPC/data frameworks, creating governance models, improving tool reliability, enabling process traceability, and translating complex shop floor needs into scalable digital and operational systems.

LEADERSHIP PROFILE

- **Ambiguity-to-system execution:** Converts unclear operational fires into structured processes, ownership models, requirements, metrics, dashboards and long-term sustainment plans.
- **Manufacturing depth plus digital leverage:** Understands the physics, equipment, process risks, operator execution, quality requirements, and data structures needed to make manufacturing systems actually work.
- **Cross-functional leadership without formal authority:** Aligns operations, engineering, IT, quality, maintenance, vendors, and leadership around practical solutions that can be adopted by users.
- **Enterprise-scope technical leadership:** Supports work beyond standard single-process engineering, including multi-site architecture, administrator enablement, governance alignment, training, requirement translation and enterprise adoption.

CAREER HIGHLIGHTS: MANUFACTURING TRANSFORMATION AND TECHNICAL LEADERSHIP

- Selected as a reference MES architect supporting multi-site adoption across Palm Bay, Tempe, Mason, and Londonderry stakeholders.
- Recovered a stalled manufacturing systems implementation and converted a paper-based production environment into a structured digital execution platform.
- Created the manufacturing systems foundation for an environment that previously had limited structure for quality, testing, qualification, SPC, preventive maintenance, metrology, defect tracking, and production monitoring.
- Built department-wide SPC and analytics framework with standards for data collection, control limits, reporting, recurring review, and process visibility.
- Designed automated environmental monitoring pipelines using Python, scheduler logic, dashboards, and Teams notifications to support controlled manufacturing operations.
- Developed a facilities manpower and workload model that directly enabled approval of hiring dedicated facilities support assigned to the department buildings.
- Reduced optical coating failures from multiple per week to near-zero within months by applying structured RCCA, equipment recovery, vendor correction, and data-driven process stabilization.
- Led ALD tool ownership at Intel D1, improving tool availability and process stability through DOE, SPC, diagnostics, pump-life analysis, and vendor coordination.
- Published 10 peer-reviewed research articles with 2,400+ citations, including work demonstrating room-temperature two-dimensional ferromagnetism in monolayer VSe_2, the first 2D magnet at room temperature.

AREAS OF EXPERTISE

Leadership and Operations	Digital transformation | Multi-site standardization | Stakeholder alignment | Technical roadmap creation | Cross-functional leadership | Change Review Board | Capacity and headcount planning | Budget and license optimization | Training and adoption | Vendor coordination
Manufacturing & Process Engineering	Process ownership | Process development | Production support | Design of Experiments (DOE) | Design for manufacturability | Tool/process qualification | Preventive maintenance strategy | Yield improvement | Traceability | Work instruction/traveler development | Operator execution support
Manufacturing Execution Systems (MES)	MES Architecture | Digital Travelers |Route definition | WIP tracking | Data Collection | SPC integration | Document Control | User roles | Functional specifications | User Acceptance Testing (UAT) support | JIRA | System enhancement workflows
Data, Reporting & Automation	Python | R | SQL | HTML | Power BI | SharePoint |VBA | Data Pipelines | Dashboards | Real-time monitoring | Metrics governance | Automated alerts
Deposition & Semiconductor Processing	MBE | ALD |E-beam evaporation |Ion Beam Sputtering | Thermal evaporation | PECVD | CVD | Photolithography | High-vacuum systems | FinFET high-k dielectric processing | Tool ownership
Optical Systems	UV-VIS and IR coatings | AR and HR coatings | Dichroic and optical filters | Thin-film modeling | Environmental and durability testing | Diamond-like Carbon (DLC) durable coatings | Recipe development | Customer drawing/specification support | TFCalc | FilmStar | OpenFilters | McLeod
Metrology & Characterization	Spectrophotometry | Ellipsometry | X-ray photoelectron spectroscopy (XPS) | Atomic Force Microscope (AFM) | Scanning Electron Microscope (SEM) | Scanning Tunneling Microscope |Physical Property Measurement System (PPMS) | Electrical characterization | JMP | OriginLab | MATLAB | LabVIEW
Quality, Risk & Failure Analysis	Failure Modes & Effects Analysis (FMEA)| RCCA | Defect taxonomy | Defect code generation | Defect classification | Process controls | Corrective action workflows | Quality data structures | Defect resolution logic
Secure / Regulated Manufacturing	Active Secret Clearance | Controlled manufacturing areas | Compliance | Documentation control | Quality governance | Safety risk assessment

PROFESSIONAL EXPERIENCE

L3Harris — Sr. Manufacturing Engineer | Palm Bay, FL **|** Mar 2024 - Present

– **Enterprise & Multi-Site Manufacturing System Leadership |** Nov 2025 - Present

- Selected as reference MES architect supporting multi-site MES adoption across Palm Bay, Tempe, Mason, and Londonderry stakeholders, with expansion alignment across Florida, Arizona, Ohio, and New Hampshire.
- Provide cross-site architecture guidance, governance alignment, administrator enablement, workflow standardization, deployment support, and best-practice sharing for manufacturing execution systems.
- Advise site leadership, engineering teams, production stakeholders, MES administrators, and project leads on system design, configuration strategy, process ownership, adoption approach, and sustainment model.
- Train future site MES administrators and functional users on workflow logic, route structures, data collection, document control, SPC concepts, user roles, troubleshooting, and governance practices.
- Serve as manufacturing-to-IT translator, converting operational needs into business requirements, functional system designs, enhancement requests, UAT scenarios, issue-tracking structures, and implementation plans.
- Drive enterprise MES licensing and alignment discussions to reduce redundant high-cost site-level solutions and support scalable enterprise adoption.

– Department Digital Transformation & Manufacturing Execution | Mar 2025 – Nov 2025

- Assumed functional ownership of a stalled MES implementation and led recovery from informal paper/Word-based travelers to digital manufacturing execution.
- Architected and deployed Eyelit MES as the foundational manufacturing execution platform for the Microelectronics Department, supporting users across operations, engineering, quality, maintenance, and leadership.
- Translated manufacturing processes into digital routes, workflows, data collection structures, user roles, process controls, WIP visibility, and document-control requirements.
- Defined MES operating model including naming conventions, operational hierarchies, ownership rules, required data collection, process-route logic, user access, documentation expectations, and long-term support approach.
- Implemented digital travelers integrating recipes, parameters, inspections, decision points, data collection, and execution logic to improve traceability and standardization.
- Designed MES functionality supporting production tracking, quality data collection, SPC, document control, tool status visibility, preventive maintenance alignment, and operational reporting.
- Developed functional documentation, training materials, process flows, and issue-tracking structures to support rollout, adoption, and sustainment.
- Owned MES enhancements, troubleshooting, testing, stakeholder feedback, and vendor coordination through JIRA and cross-functional working sessions.
- **Awarded the Rising Star Award** for delivering high-impact manufacturing technology solutions, including Eyelit MES implementation and automated visualization platforms.

– Manufacturing Systems Foundations, Process Governance & Operations Support | Mar 2024 – Mar 2025

- Built process governance framework connecting quality systems, preventive maintenance, metrology, defect classification, failure analysis, production monitoring, and operational decision-making.
- Designed and institutionalized the department's first end-to-end SPC and analytics framework, authoring plans, standards, review cadence, metrics governance, and user training.
- Developed dashboards for tracking tools and automated notifications.
- Implemented environmental health monitoring strategy with RH/temperature logger deployment, ISO-based alarm limits, automated Python pipelines, live dashboards, and Teams notifications.
- Served on the Change Review Board, evaluating process changes and partnering with deposition engineering to improve e-beam chamber performance, including revised crystal monitoring methods.
- Developed department facilities workload model by analyzing facility-level assets, reviewing manuals, extracting preventive maintenance tasks, validating labor hours/frequencies, and building a year-long two-shift task calendar using utilization and non-productive-time assumptions to justify dedicated facilities headcount, resulting in approved budget support.
- Developed failure-mode and defect-classification framework for microelectronics manufacturing, including defect codes, failure categories, and resolution logic.
- Supported safety risk assessments, operational readiness activities, documentation, and compliance efforts in a controlled, regulated manufacturing environment.
- **Awarded two R.I.S.E. Awards** for preventive maintenance strategy and safety risk assessment contributions.

Lockheed Martin— Sr. Optical Manufacturing Process Engineer | Orlando, FL | Jan 2022 – Jan 2024

- Served as coating-area subject matter expert for Lockheed Martin Missiles and Fire Control, providing production support to approximately 8 process engineers and 30 operators across optical coating production, process engineering, operator execution, equipment readiness and coating design.
- Supported major defense programs including F-35, Sniper Advanced Targeting Pod (ATP), Javelin, Joint Air-to-Ground Missile (JAGM), Modernized Day Sensor Assembly (MDSA), IRAD initiatives and additional non-recurring programs.
- Led thin-film coating production for antireflective, high-reflective and environmentally durable optical coatings across visible and infrared optical systems.
- Owned full lifecycle coating development for customer-requested, IRAD, and NRE work, including coating design, optical modeling, project budgeting, material selection and purchasing, equipment setup, recipe development, pass/fail requirement definition, production execution, and shipment support.
- Designed and developed coating processes in accordance with customer drawings, program specifications, MIL-PRF-13830 requirements where applicable, and internal production controls.
- Served as manufacturing owner for PVD and CVD deposition systems from Tecport Optics and Bühler Leybold, including e-beam evaporation, sputtering, thermal evaporation, and plasma-enhanced CVD.
- Diagnosed and resolved complex coating and equipment issues impacting yield, durability, and reliability, including coating delamination, non-uniformity, spectral deviation, environmental durability failures, chamber leaks, witness sample absorption, and temperature calibration errors.

- Implemented structured RCCA methods that significantly reduced defective coating runs and enabled production delivery up to one month ahead of schedule for select programs.
- Recovered critical manufacturing assets previously labeled out-of-production, including an environmental humidity test chamber and PECVD system for diamond-like carbon coatings.
- Identified supplier and vendor quality gaps contributing to manufacturing failures and led corrective actions to align equipment, materials, and supplier capability with program requirements.
- **Awarded 4 Next Gen Recognition Awards** for cross-functional collaboration, yield improvement, root-cause resolution, and operational performance impact.
- **Performance Management Team (PMT) Award**: Leadership in resolving operator issues and improving coordination.

Intel Corporation — Semiconductor Process Engineer | Hillsboro, OR | Jun 2020 – Jul 2021

- Supported high-volume front-end semiconductor manufacturing at Intel's D1 fab as ALD tool owner, driving equipment reliability, film quality, process stability, troubleshooting, and technology transition support.
- Owned two front-end ALD manufacturing tools comprising four deposition sub-chambers supporting high-k dielectric film deposition for FinFET transistor production.
- Improved tool availability and process stability using DOE, SPC, chamber pressure diagnostics, preventive maintenance optimization, and pump-life analysis.
- Coordinated directly with equipment vendors to sustain high-volume production and resolve equipment performance issues.
- Led conversion of two tools from a previous product line to a current production flow, supporting manufacturing continuity during technology transitions.
- Restored inactive tools to production by coordinating diagnostics, vendor support, chamber recovery, and production qualification activities.

RESEARCH & TECHNICAL EXPERIENCE

University of South Florida — Research Associate: Advanced Materials Physics

Tampa, FL | Aug 2013 – Jun 2020

- Led research enabling the first reported room-temperature ferromagnetism in monolayer VSe_2, opening a new area of two-dimensional magnetism research.
- Work was highlighted by Nature Nanotechnology editorial coverage, "2D Magnetism Gets Hot," as expanding possibilities for next-generation device engineering.
- Synthesized and characterized monolayer VSe_2 and developed deposition recipes for TiSe2, MoSe2, WSe2, and TaSe2 to study magnetic states, charge density waves, excitonic behavior, band structure, doping, and surface morphology.
- Specialized in physical vapor deposition using Molecular Beam Epitaxy and ultra-high-vacuum experimental platforms.
- Installed, configured, maintained, and commissioned ultra-high-vacuum systems, including a shared departmental MBE deposition system supporting multi-user research operations.
- Designed end-to-end experimental workflows and data-analysis systems for characterizing two-dimensional materials.
- Built predictive models linking process parameters to electronic, magnetic, structural, and surface behavior in advanced materials systems.
- Selected as guest scientist at SLAC National Accelerator Laboratory in California and SOLEIL Synchrotron in France for advanced characterization studies.

University of Puerto Rico — Research Associate: Nanomaterials, Polymer Electronics & Sensor Devices

Humacao, PR | Aug 2010 – Aug 2013

- Conducted hands-on research in polymer electronics, nanofiber sensing devices, microfabrication, electrical characterization and defense-related smart-textile concepts through PREM collaborations with the University of Pennsylvania and University of Nebraska-Lincoln.
- Fabricated gas-sensing diodes and transistors by electrospinning polymer nanofibers directly onto silicon wafers to form active sensing layers for chemical detection.
- Performed metal electrode deposition via thermal evaporation to define device contacts and circuit structures.
- Executed electrical characterization, including I-V measurements and gas response testing, to evaluate device sensitivity, stability, and performance.
- Conducted structural and morphological analysis using SEM to validate nanofiber formation and device architecture.
- Developed and optimized polymer formulations and nanofiber recipes to tune chemical composition and improve gas-sensing response.

- Supported smart-textile concepts integrating sensing devices into fabrics for environmental and defense-related gas detection.
- Contributed to carbon-nanotube-based sensing platforms, including early non-invasive glucose detection device development using CVD and photolithography methods.

AWARDS

- L3Harris Rising Star Award - recognized for MES implementation, manufacturing technology solutions, automated visualization platforms, innovation, cross-functional collaboration, and sustainable long-term adoption.
- L3Harris R.I.S.E. Awards (2) - recognized for preventive maintenance strategy and safety risk assessment contributions.
- Lockheed Martin Next Gen Recognition Awards (4) - recognized for cross-functional collaboration, yield improvement, root-cause resolution, and operational performance impact.
- Lockheed Martin Performance Management Team Advocate Award - recognized for improving operator-management communication and daily issue-resolution workflows.

EDUCATION & ACADEMIC RECOGNITION

- **Ph.D., Physics** | University of South Florida (USF)
- **M.Sc., Physics** | University of South Florida (USF)
- **B.Sc., Physics Applied to Electrical Engineering** | University of Puerto Rico (UPR)
- **Near-minor-level coursework in Computational Mathematics** | University of Puerto Rico (UPR)
- **Florida McKnight / FMLSP Graduate Fellowship Recipient**
- **APS Bridge Program Physics Graduate Fellow — Inaugural Cohort** | University of South Florida (USF)
  Featured by American Physical Society (APS) News as a Bridge Program PhD alumnus.

CERTIFICATIONS & CONTINUING EDUCATION

- **Google Data Analytics Professional Certificate** | Google via Coursera
- **Leading People and Teams Certificate — In Progress** | University of Michigan via Coursera
- **CS50 Computer Science Coursework — In Progress** | Harvard University via edX

TEACHING & TRAINING

- **Graduate Teaching Associate** | Physics Department, University of South Florida
  Developed and delivered laboratory instruction for Physics I and Physics II courses, including experiment setup, student guidance, technical explanation, grading, and laboratory safety expectations.
- **Academic Tutor** | Mathematics and Physics, University of Puerto Rico
  Tutored undergraduate students in Calculus I–II, Precalculus I–II, and Physics I–II through university library tutoring services.

RESEARCH IMPACT & EXTERNAL SCIENTIFIC EXPERIENCE

- 10 peer-reviewed publications with 2,400+ citations | Google Scholar Profile
- Guest Scientist: SLAC National Accelerator Laboratory, California, USA.
- Guest Scientist: SOLEIL Synchrotron, Paris, France.
- Research Internship / PREM Collaborations: University of Pennsylvania and University of Nebraska-Lincoln.